

02 / May / 2026

Agenda:

Feature Engineering-5

Class:

Time → Wednesday class alternate

~~Extend~~

8th May - no Wednesday class  
FST

### (1) Min max scaling (Normalization)

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \rightarrow [0, 1]$$

Age  $\rightarrow$   $\begin{array}{ccc} 26 & \longleftrightarrow & 81 \\ \text{min} & & \text{max} \\ \downarrow & & \downarrow \\ 0 & & 1 \end{array}$

$$\frac{26 - 26}{81 - 26} = \frac{0}{55} = \underline{0}$$
$$\frac{81 - 26}{55} = \frac{55}{55} = \underline{1}$$

$\rightarrow$  sensitive to outliers

### (2) Standardization (z score scaling)

$$x' = \frac{x - \mu}{\sigma}$$

$$\text{mean} = 0, \text{std} = 1$$

Handles outliers better than min-max scaling

### (3) Robust scaler (Median & IQR based)

$$x' = \frac{x - \text{median}(x)}{\text{IQR}(x)}$$

Robust to outliers

### (4) log Transformation

$$x' = \log(1 + x)$$

$\rightarrow$  reduces right skewness

Note: (For left skewness, we use:)

$\rightarrow$  square root / cube root  
 $\rightarrow$  Box-Cox (positive data)

$\rightarrow$  Yeo-Johnson (works with negative too)

$$x' = \sqrt{x}$$

$$x' = x^{1/3}$$

$$\text{Box-Cox } x \longrightarrow x' = \frac{(x^\lambda - 1)}{\lambda} \longrightarrow \lambda \neq 0$$

$$\log(x) \longrightarrow \lambda = 0$$

Yeo-Johnson (generalization of Box-Cox)

→ works with both +ve/-ve values

$$\begin{aligned} &\rightarrow x > 0 \\ &x < 0 \end{aligned}$$

Reciprocal transformation:

$$\rightarrow x' = \frac{1}{x}$$

Scaling  $\longrightarrow$  removing unit  
 $\downarrow$  way  
 fixing dist

$\neq$  remove the outliers

scaling  $\neq$  fixing the distribution

→ goal: to remove unit

: make features on the same scale, to remove dominance

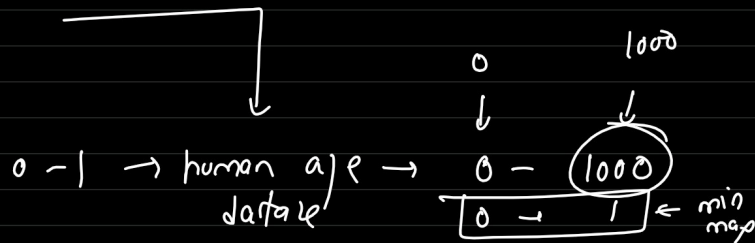
Practical example

0-100  $\xrightarrow{\text{min-max}}$

human  $\rightarrow 10 \rightarrow 0.1$

human  $\rightarrow 90 \rightarrow 0.9$

human  $\rightarrow 100 \rightarrow 1.0$



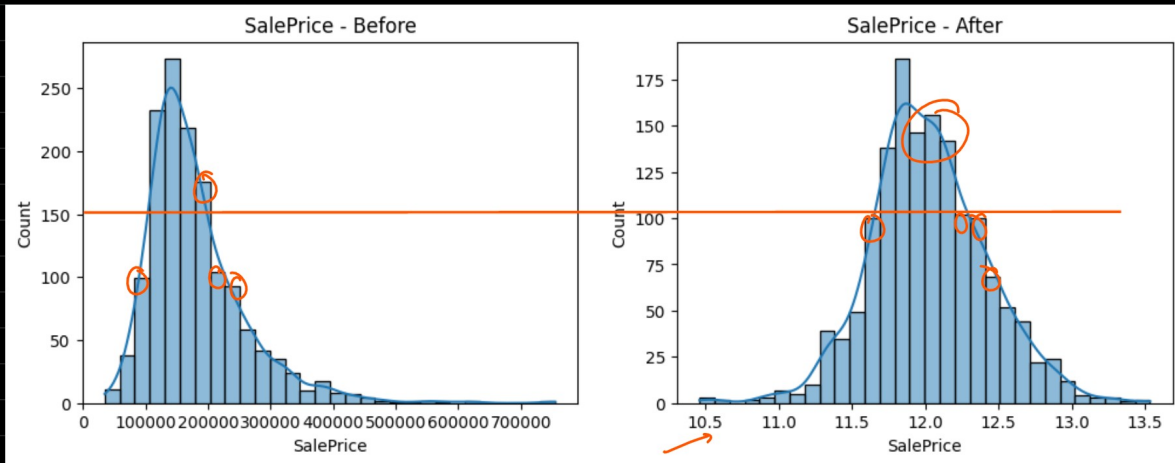
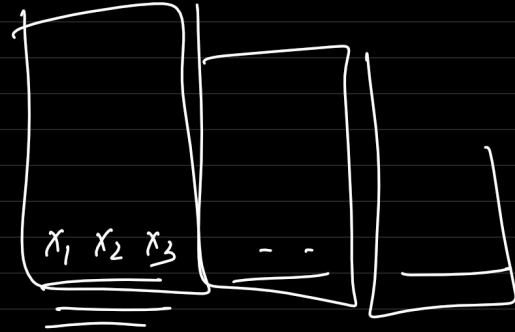
human  $\rightarrow 0-100$

$\downarrow$   
 $\boxed{0-0.1} \quad \boxed{1}$

$\boxed{>0.1} \quad \boxed{<1}$



10, 11, 12, 13, 14, 15, ... 20  
 \_\_\_\_\_  
 bin

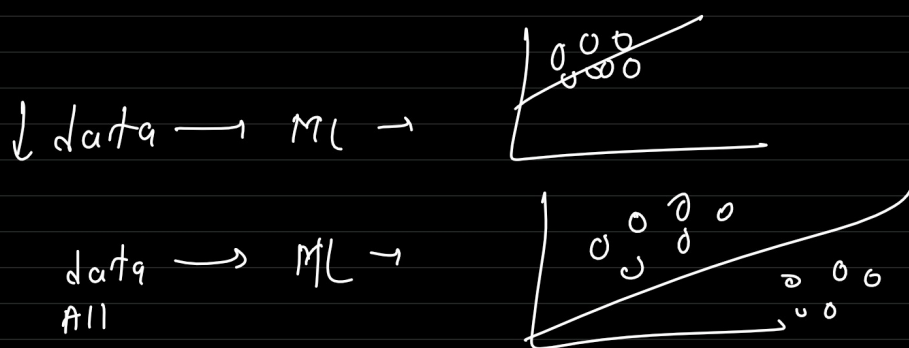
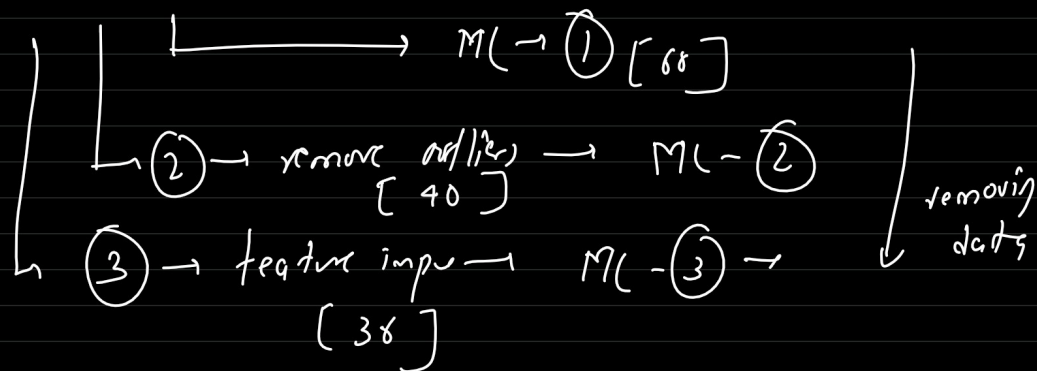


Ques: why to apply scaling in the feature engineering process.

- remove outlier → z-score →
- modified-zscore →  → filter →
- handle the null value,
- feature selection →

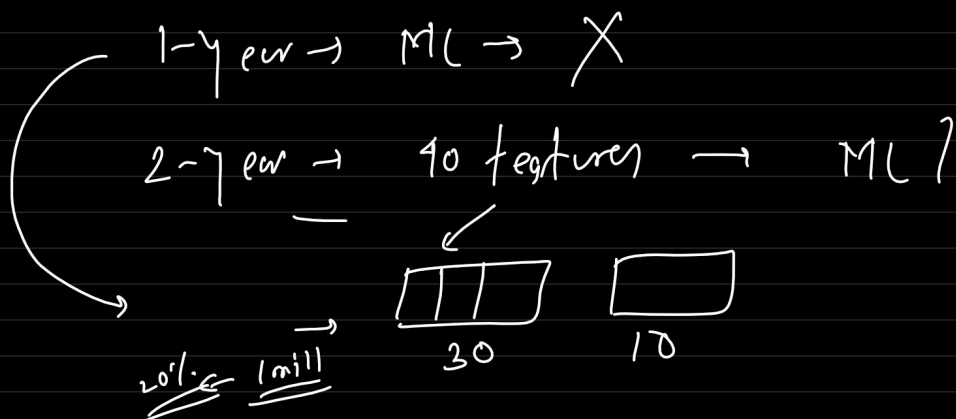
→ 1(a) → remove unit  
→ 1(c) readable

→ fixing the distribution



1-year → 30 features → 1 mill

2-year → 40 features → 10 features  
→ 1 mill



primary technique for left-skewed data:

(1)  $x^2$  (square transformation)  $\rightarrow$  Assignment  
 $\rightarrow$  Assignment

(2)  $x^3 \dots x^5$

(3) Reflection (reflect & transform)  $\rightarrow$  Assignment

$\rightarrow$  reflect

$\rightarrow x' = \text{constant} - x$

$\rightarrow$  left  $\rightarrow$  right skew

$\downarrow$   
log, square-root

(max + 1)

(4) Yeo-Johnson

2, 3, 4, 50, 40

1, 2, 3, 4, 5, 100, 101, 1000, 1001

1  $\rightarrow$  50

100  $\rightarrow$  101

1000  $\rightarrow$  1200

70%

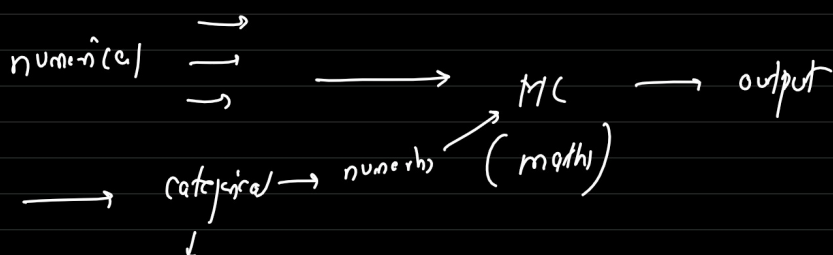
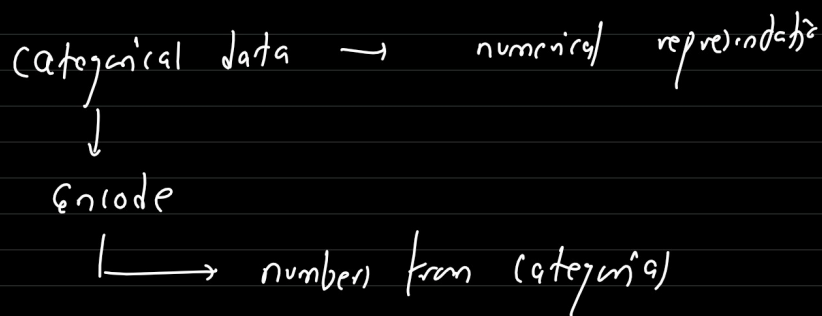
20%

10%

small numbers =  $\left[ \begin{array}{l} 2 \rightarrow 2^2 \rightarrow 4 \\ 3 \rightarrow 3^2 \rightarrow 9 \end{array} \right]^5$

large numbers =  $\left[ \begin{array}{l} 1000 \rightarrow 1000^2 \\ 1001 \rightarrow 1001^2 \end{array} \right]$  gap is very high

# Data Encoding technique:



nominal, ordinal

$\downarrow$   $\downarrow$

categories with no order categories with order

column  $\rightarrow$  categories

(A, B, C) (A, B, C)

$\downarrow$

0, 1, 2

(cat. code)

| color      | frequency     |
|------------|---------------|
| (1) Red    | 400 - 480 THz |
| (2) orange | 480 - 510 THz |
| (3) yellow | 510 - 530 THz |
| (4) Green  | 530 - 600 THz |
| (5) Blue   | 600 - 670 THz |
| (6) Indigo | 670 - 700 THz |
| (7) violet | 700 - 750 THz |

red, blue, indigo, orange, violet

|        |   |
|--------|---|
| red    | 0 |
| orange | 1 |
| blue   | 2 |
| Indi   | 3 |
| violet | 4 |

| color | freq | category of color | LE | LE | B_OHE | R_OHE | G_OHE | V_OHE | O_OHE |
|-------|------|-------------------|----|----|-------|-------|-------|-------|-------|
| 625   |      | Blue              | 0  | 2  | 1     | 0     | 0     | 0     | 0     |
| 420   |      | Red               | 3  | 0  | 0     | 1     | 0     | 0     | 0     |
| 680   |      | Indigo            | 1  | 3  | 0     | 0     | 1     | 0     | 0     |
| 690   |      | Indigo            | 1  | 3  | 0     | 0     | 1     | 0     | 0     |
| 710   |      | violet            | 4  | 1  | 0     | 0     | 0     | 1     | 0     |
| 490   |      | orange            | 2  | 1  | 0     | 0     | 0     | 0     | 1     |

Label Encoder → no order → 0, 1, 2 ...  
 (nominal) → (alphabetical)

Label Encoder → order  
 (ordinal) → fact as order

Male → 0  
 Female → order? 1  
 other → nominal → 2

← magnitude  
 ↓  
 1

|        | Male | Female | other |
|--------|------|--------|-------|
| Male → | 1    | 0      | 0     |
| Female | 0    | 1      | 0     |
| other  | 0    | 0      | 1     |

Male → [ 1 0 0 ] → one hot encoding  
 Female → [ 0 1 0 ]  
 other → [ 0 0 1 ]  
 ↓  
 OHE



## Frequency Encoding

| color | freq | category of color | freq-encoding |
|-------|------|-------------------|---------------|
| 625   |      | Blue              | 1             |
| 420   |      | Red               | 1             |
| 680   |      | Indigo            | 2             |
| 690   |      | Indigo            | 2             |
| 710   |      | violet            | 1             |
| 490   |      | orange            | 1             |

give more priority to highly repeated values

## Target Encoding

← Target (7)

| color | freq | category of color | Target-Encoding |
|-------|------|-------------------|-----------------|
| 625   |      | Blue              | 625             |
| 420   |      | Red               | 420             |
| 680   |      | Indigo            | 685             |
| 690   |      | Indigo            | 685             |
| 710   |      | violet            | 710             |
| 490   |      | orange            | 490             |

Blue → 625

Red → 420

Indigo →  $\frac{680 + 690}{2}$   
(mean)

violet → 710

orange → 490

1 mil

n-unique  $\approx$  rows